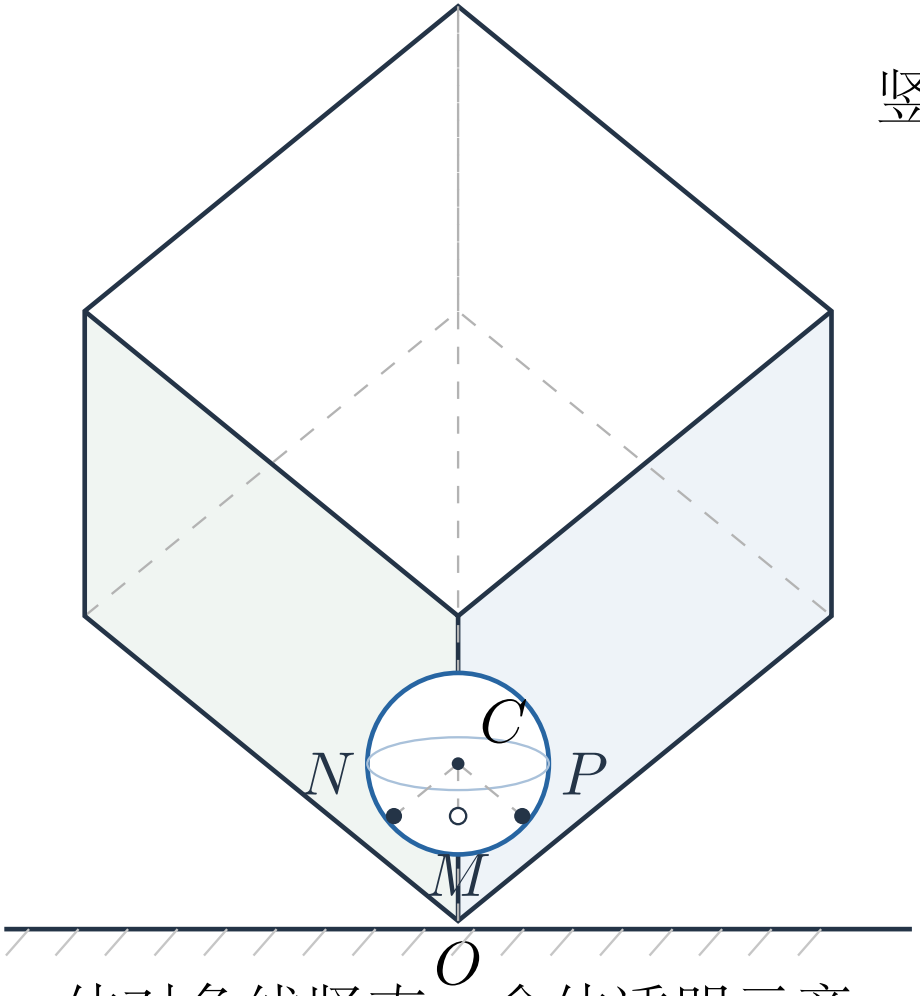
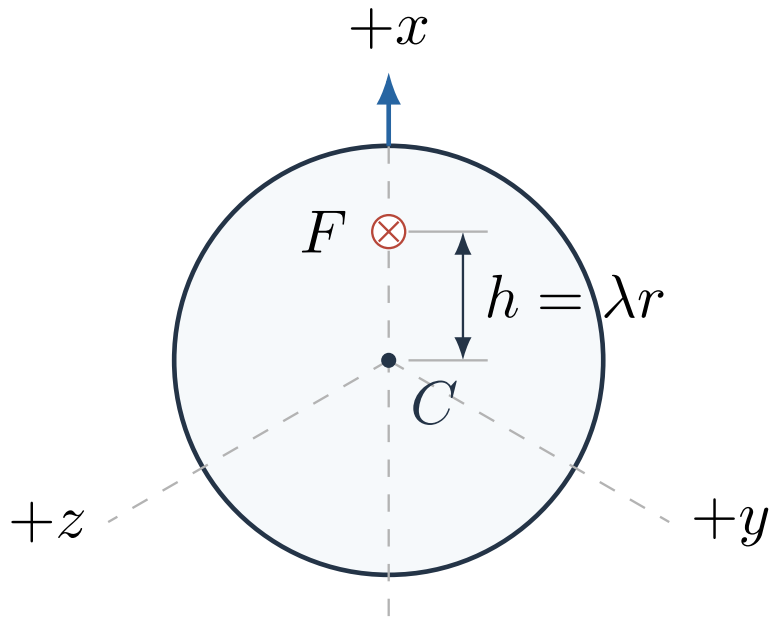
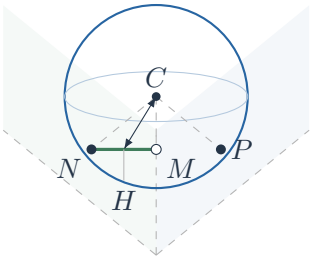


体对角线竖直；盒体透明示意

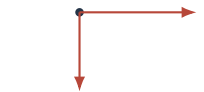


俯视：F 竖直向下，投影为入纸面



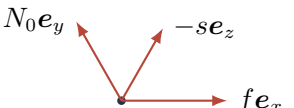
H 为 MN 中点，$CH = r/\sqrt{2}$

P 点



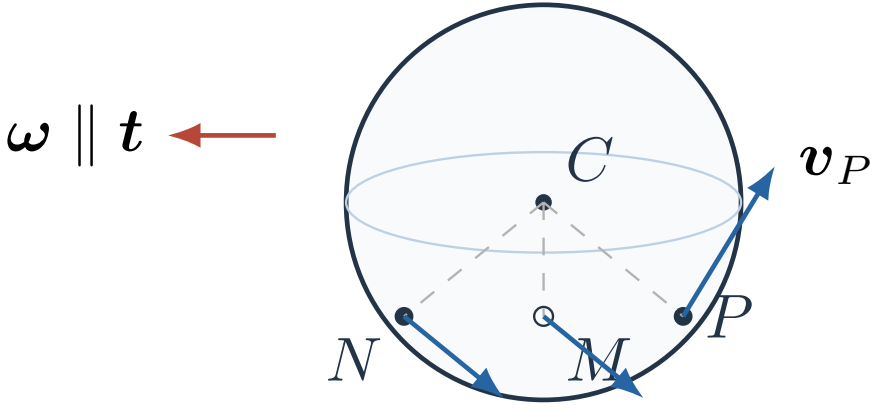
M 点

N 点由 y, z 互换



$|\boldsymbol{f}_M| = \sqrt{f^2 + s^2}$

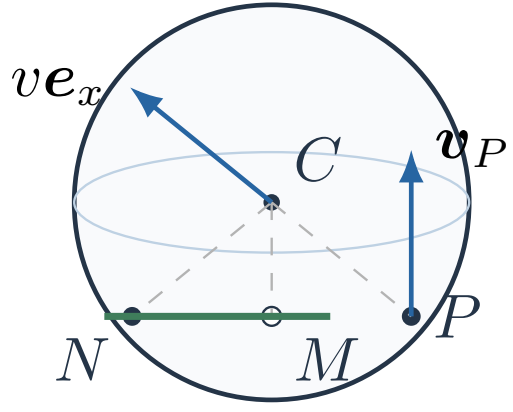
固定球心打滑



$\boldsymbol{v}_M, \boldsymbol{v}_N \parallel -\boldsymbol{e}_x$

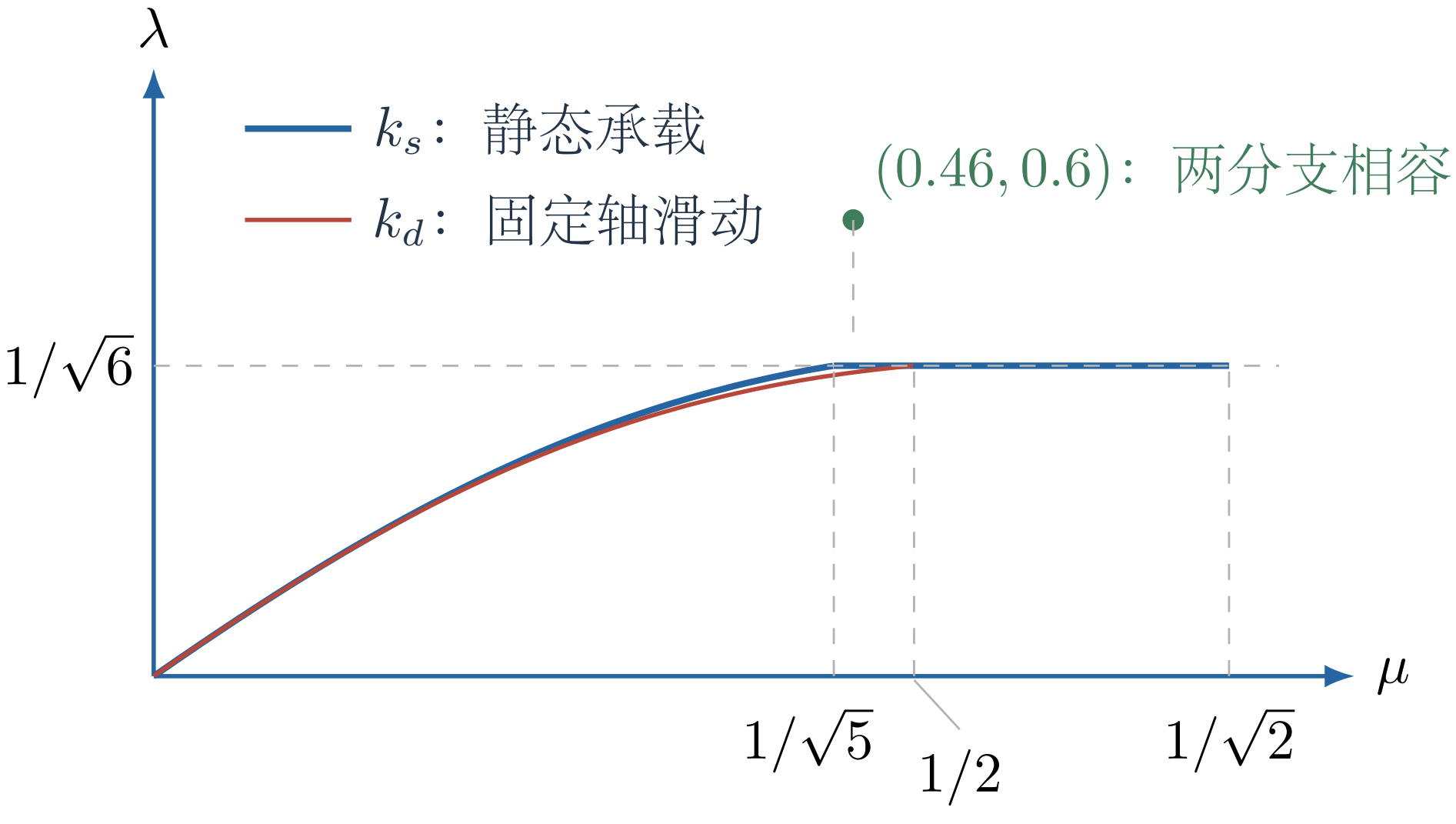
$\boldsymbol{v}_C = 0$，摩擦与各点滑速反向

两壁纯滚动



$\boldsymbol{v}_M = \boldsymbol{v}_N = 0$

$|\boldsymbol{\omega}| = \sqrt{2}v/r$，$P$ 接触分离



λ 超过相应曲线，才可能在有限载荷到达该门槛

$\mu > 1/\sqrt{2}$ 另须检查静力自锁；未画成“必滚区”